

TECHNICAL ASSESSMENT REPORT

Bangkok Airbnb Market Intelligence

An End-to-End Data Engineering, Statistical, and Machine Learning Analysis

Inside Airbnb Real World Data Challenge

Submitted to

Expernetic (Pvt) Ltd - Data Engineer Intern Talent Assessment Program

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July 2026

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1. Executive Summary

This report presents an end-to-end analysis of 28,697 Bangkok Airbnb listings, drawn from the Inside Airbnb public dataset. The work spans data engineering (ingestion, cleaning, dimensional modeling), exploratory analysis, formal statistical hypothesis testing, machine learning (price prediction and segmentation), and applied AI/NLP with all prices reported in Thai Baht (THB).

Key Findings

- Median listing price is 1,578 THB/night; the top 1% of hosts (79 hosts) control 21.8% of total supply (6,781 listings), this market includes significant commercial-scale operators, not just individual hosts.
- After removing 290 listings verified as non-genuine placeholder prices, neighbourhood explains a medium share of price variation ($\eta^2 = 0.0612$), and listings farther from the city centre are modestly cheaper ($r = -0.153$).
- Regression analysis shows guest capacity, bedrooms, and bathrooms are the strongest price drivers, together explaining 36–45% of price variance depending on model specification.
- Several hypotheses showed statistically significant price differences ($p < 0.001$) but negligible practical effect sizes — a consequence of large sample size, not strong relationships. Host reputation (Superhost status) was the one exception, showing both significance and a meaningful effect (Cohen's $d = 0.562$) on review scores.
- An XGBoost model predicts price with a Test MAE of 702.64 THB and MAPE of 41.9%, outperforming a linear baseline by 21.2%. The model is measurably less reliable for Hotel room listings due to smaller sample size.
- K-Means clustering identified six behaviorally distinct host/listing segments, from Professional Multi-Property Operators to a confirmed genuine at-risk segment with poor ratings.
- Star ratings are heavily compressed (59.5% of listings rated 4.8+) and only weakly correlated with review text sentiment ($r = 0.238$) the numeric score alone is a weak quality signal in this market.

Actionable Recommendations

- Hosts should prioritize listing capacity (accommodates, bedrooms, bathrooms) over amenities count or review count. These are the strongest, most reliable levers at an achievable price.
- Entire home/apt and Private room are different economic categories (median ~2,406 THB vs. ~1,902 THB) and should be chosen deliberately based on target guest segment and capital, not treated as interchangeable.
- Platform operators should adopt a systematic flag-and-verify approach to data quality (as demonstrated in this report's data quality investigation) rather than relying on simple percentile-based outlier filtering.
- Investors should think in terms of the six identified segments rather than city-wide averages as high-occupancy and high-price/low-volume represent two different, equally valid strategies.
- Given confirmed rating compression, scores below 4.7 should be treated as a materially stronger warning sign than they appear at face value.

Notable Limitations

- 290 listings (~1%) were excluded as verified non-genuine prices disclosed and documented, not silently dropped.
- Sentiment analysis was performed on a 5,000 review sample using a general-purpose multilingual model. Manual validation identified occasional misclassifications, particularly for short or context dependent reviews, and these limitations were considered when interpreting the results.

- This analysis covers a single city and a single data snapshot; findings may not generalize to other markets or time periods without further validation.

2. Objectives & Scope

The brief for this assignment explicitly states that it is scoped beyond what can reasonably be completed in the allotted time, and that candidates are evaluated on prioritization, depth, and communication rather than completion of every listed task. This report follows that principle directly: fewer sections were pursued, each to a professional standard of depth, rather than shallow coverage of the entire brief.

City Selection

Bangkok was selected as the analysis city. At 31,069 raw listings, it offers a dataset large enough to support meaningful statistical testing (ANOVA across 45 neighbourhoods, regression, clustering) while remaining tractable within the assignment's one-week timeframe. A single-city scope was chosen deliberately over a multi-city comparison (see Section 12) to allow full analytical depth on one market rather than partial coverage of several.

Scope Decisions

Assignment Area	Decision	Rationale
Data Engineering (ingestion, cleaning, modeling)	Full depth	Mandatory; foundation for all downstream work
EDA (Sections 4.1–4.3)	Full depth	Recommended; core analytical thinking demonstration
Statistical Analysis (5.1–5.3)	Full depth	Recommended; includes a full data-quality-driven correction cycle
ML - Price Prediction & Segmentation (6.1, 6.3)	Full depth	High business value; reused directly in the dashboard
ML - Demand Forecasting (6.2)	Deferred	Deferred due to lower priority and high time investment relative to expected impact: noted in Section 14
ML - Full Cross-City Bias Testing (6.4)	Partially folded into 6.1	Residual bias-by-segment analysis performed; true cross-city transfer requires a second city's full pipeline
AI/NLP - Sentiment + LLM Summary (7.1, 7.2)	Full but lightweight	Demonstrates AI literacy and critical evaluation without over-investing in an optional, lower-weighted section
Cloud-Native / Open Innovation (3.6, Section 8)	Descope in favor of an interactive dashboard	Expert tier and optional; a working Streamlit dashboard were judged higher value given time constraints

Prioritization principle

Every deferral above is disclosed here and revisited in Section 14 (Future Improvements) - none are silent omissions. This mirrors the assignment's own evaluation note that thoughtful prioritization and honest scoping are themselves scored criteria, separate from raw task completion.

3. Dataset Overview

All data originates from Inside Airbnb, a publicly available, independent dataset providing detailed Airbnb listing information for cities worldwide. No supplementary or external data sources were used.

Files Used

File	Rows	Description
listings.csv.gz	31,069	Core listing data: price, room type, location, host info, amenities
calendar.csv.gz	11,340,155	Daily availability per listing (365-day window from scrape date)
reviews.csv.gz	693,558	Guest review text with reviewer ID, date, listing reference
neighbourhoods.csv	50	Neighbourhood names and geospatial groupings

Schema & Relationships

listings.id is the primary key, referenced as listing_id (foreign key) in both calendar.csv.gz and reviews.csv.gz. neighbourhood_cleansed in listings joins to neighbourhoods.csv on neighbourhood name. This forms the basis of the star schema detailed in Section 5.

Data Quality Observations at Profiling Stage

- 14 columns were found to be 100% null in this scrape, including host_since, host_total_listings_count, host_response_rate, and license. These were dropped during cleaning rather than imputed, since there is no information to recover.
- price is stored as text with a currency-style '\$' prefix; verified against raw source magnitudes to confirm the underlying values are genuinely Thai Baht (THB), not USD. The '\$' is a generic template character in Inside Airbnb's export format, not a currency indicator.
- Boolean-style fields (host_is_superhost, host_has_profile_pic, host_identity_verified) are stored as 't'/'f' strings rather than native booleans.
- review_scores_rating is missing for ~33% of listings and verified to correlate directly with number_of_reviews = 0 (i.e., genuinely new, unreviewed listings), not random missingness.

Assumptions Documented

- neighbourhood_cleansed (0% null) was used as the canonical location field, since neighbourhood_group_cleansed is entirely null in this scrape.
- calculated_host_listings_count (0% null) was used for host portfolio size rather than host_total_listings_count, which is 100% null.
- The calendar file for this scrape contains no daily price column and therefore revenue estimated as price × estimated_occupancy_l365d (from listings.csv) rather than derived directly from calendar data.
- Duplicate detection was performed on an exact-match and key-match basis (listings.id, reviews.id); no exact duplicate rows were found in listings.csv.gz at the profiling stage.
- Geographic fields (latitude, longitude) were validated against domain bounds (−90 to 90, −180 to 180) and zero invalid coordinates were found.

Business Domain Context

Each entity in this dataset represents a distinct actor in a two-sided short-term rental marketplace. A listing is a bookable unit (entire apartment, private room, shared room, or hotel-style unit) offered by a host, who may operate a single listing casually or, as this analysis repeatedly finds, dozens of listings as a commercial operation. A review represents a completed guest stay and is the platform's primary trust signal, subject to the rating-inflation dynamics documented in Section 6. A neighbourhood is a geographic grouping used both for host-facing search and for this analysis's location-based aggregation.

Single City Scope - Multi-City Guidance Considered

The assignment's City Selection Guide notes that candidates selecting multiple cities should additionally compare dataset sizes, recency, and coverage, and document a harmonization strategy before engineering begins. This guidance was reviewed and consciously not applied here, since a single-city, full-depth approach was judged the stronger use of available time (see Section 2). The pipeline's config-driven design (Section 5) means this harmonization step would primarily involve verifying consistent column names and scrape dates across cities a natural next step documented in Section 14.

4. Methodology

The overall analytical approach followed a deliberate sequence: profile before cleaning, clean before modeling, model before analyzing, and critically verify before concluding. This last principle was applied repeatedly and is the throughline of the entire project (see Section 8, Data Quality Investigation).

Method Selection Reasoning

- DuckDB was chosen for calendar aggregation (11.3M rows) overloading the full file into pandas - DuckDB queries the compressed file directly without exhausting memory, appropriate for a file of this scale.
- Non-parametric tests (Mann-Whitney U) were used in place of t-tests for price comparisons, since price is right-skewed and violates the normality assumption a t-test requires.
- Effect sizes (Cohen's d, eta-squared) are reported alongside every p-value throughout this report with a sample size in the tens of thousands, statistical significance alone is close to guaranteed and does not indicate practical importance.
- Log-transformation of price was applied for regression and price-prediction modeling after diagnosing severe right-skew and non-normal residuals in an initial raw-price model which is a standard, defensible correction rather than an arbitrary choice.
- Median imputation (not deletion) was used for missing review scores in modeling, paired with a retained `has_reviews` flag. This preserves the ability to price brand new, unreviewed listings, arguably the most realistic use case for a price-prediction tool.

Validation Discipline

Every major finding in this report was checked against at least one of: raw source data, an alternative model specification, or a manual read of underlying text/records. This is not a formality. It directly produced the single most consequential finding in the project (Section 8) and caught two distinct AI-tool failure modes (Section 9).

Technical Stack

Category	Tools Used
Languages	Python 3, SQL
Data Processing	pandas, DuckDB
Statistics	scipy, statsmodels
Machine Learning	scikit-learn, XGBoost, SHAP
NLP	VADER (NLTK), Hugging Face Transformers
Visualization	matplotlib, seaborn, Plotly, Folium
Dashboard	Streamlit
Database	DuckDB (local analytical warehouse)
Version Control	Git

This stack was selected for appropriateness to a single-city, time-boxed assignment rather than for production scale. DuckDB in particular was chosen specifically because it allows querying an 11.3 million row compressed CSV directly, without either loading the full file into memory or standing up a separate database server.

5. Engineering Approach

Pipeline Architecture

Pipeline Flow

Ingest (retry + logging) → Clean & Flag → Process Calendar (DuckDB) → Enrich → Build Star Schema → EDA → Statistics → ML → AI/NLP → Dashboard

The pipeline is fully config-driven via a single config.yaml file. The target city, source URLs, and processing parameters are all externalized, so the entire pipeline can be re-pointed at a different city with no code changes. A single command (run_pipeline.py) executes every stage in order and halts on first failure, rather than silently continuing with partial data.

Repository Structure

- src/ingestion/ :- download pipeline with retry logic, structured logging, and per-file success/failure reporting
- src/processing/ :- cleaning, calendar aggregation, enrichment
- src/modeling/ :- star schema construction in DuckDB
- src/analysis/{eda, statistics, datascience, ai}/ :- grouped by shared data and analytical logic rather than by assignment section number, to avoid duplicating query/loading code across scripts
- src/utils/config.py :- a single shared configuration loader used by every script

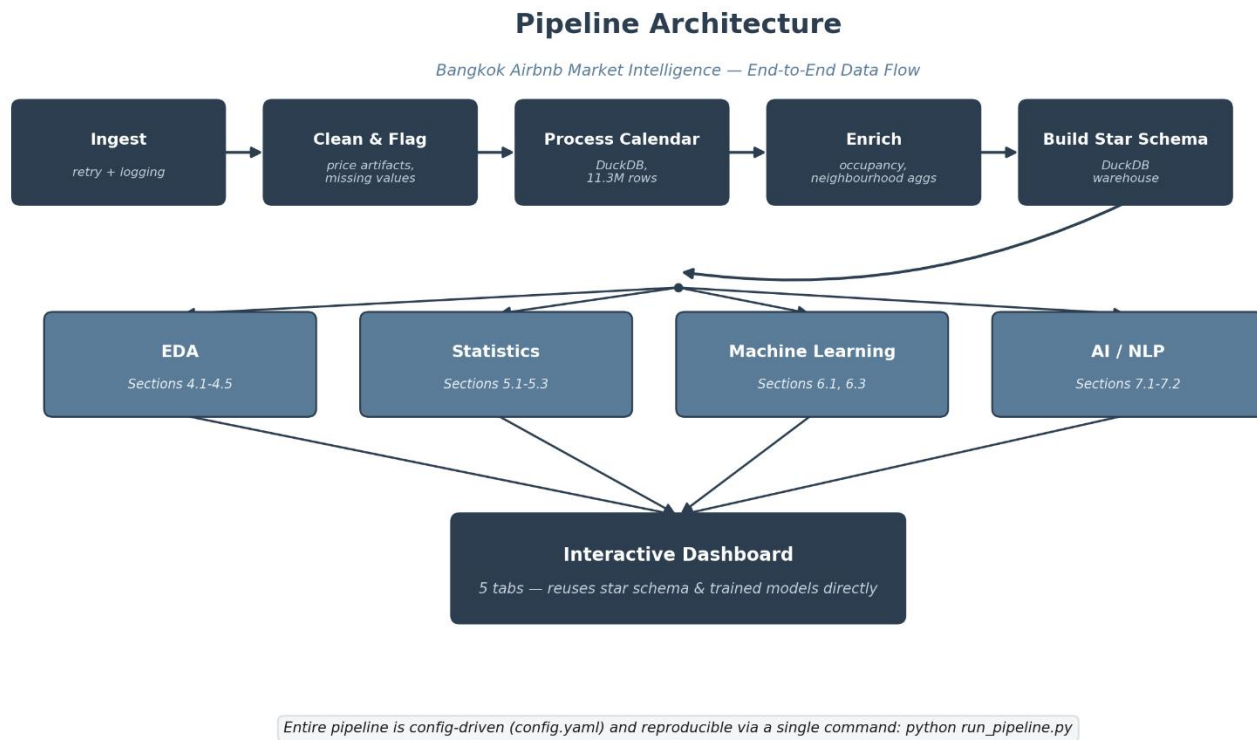


Figure 01 Pipeline Architecture: config-driven ingestion through dashboard, with parallel analysis branches sharing a single star schema.

Data Model - Star Schema

A dimensional model was implemented in DuckDB with one fact table at listing grain and three dimension tables:

Table	Grain / Role	Key Columns
fact_listings	One row per listing	price, price_likely_artifact, occupancy_rate_pct, review metrics
dim_host	One row per host	host_id, host_is_superhost, calculated_host_listings_count
dim_property	One row per listing	room_type, property_type, accommodates, bedrooms, bathrooms, beds
dim_neighbourhood	One row per neighbourhood	neighbourhood_median_price, neighbourhood_listing_count, avg rating

Engineering Decision - Fact Table Grain

A listing-grain fact table was chosen over a daily-grain table. This is a single-scrape snapshot, not longitudinal transactional data therefore, a date dimension would be largely redundant. Trade-off: day-by-day occupancy analysis is not directly supported at the fact-table level; occupancy is instead pre-aggregated into `occupancy_rate_pct` during enrichment, which is appropriate for this dataset's actual grain.

Star Schema — Data Model

One fact table at listing grain, three dimension tables



Fact table grain: one row per listing (single-scrape snapshot, not longitudinal transactional data)

Figure 02 Star Schema: A listing level fact table linked to three dimensions

Reproducibility & Production Considerations

- All intermediate outputs (cleaned data, star schema, trained models) are cached to disk as Parquet/DuckDB/joblib files, so downstream scripts do not repeat upstream work.
- Every pipeline run produces a timestamped log file, providing an audit trail of ingestion attempts, retries, and failures.
- Disclosed limitation: the current pipeline is not idempotent in the incremental-processing sense, and it reprocesses all data on every run rather than detecting and skipping unchanged inputs. This is acceptable for a single, bounded assignment but would need a checkpointing strategy for continuous production use.
- Disclosed limitation: the DuckDB warehouse is a single local file, appropriate for this assignment's scope but not for concurrent multi-user access; a production deployment would use a managed warehouse (e.g. BigQuery, Snowflake, or PostgreSQL).

6. EDA Findings

6.1 Summary Statistics & Distributions

Mean price is 2,222.22 THB (post-correction; see Section 8), with the distribution remaining right-skewed even after removing non-genuine prices — the median is a more reliable reference point than the mean for individual pricing guidance.

- Top 1% of hosts (79 hosts) control 21.8% of all listings (6,781 listings), highlighting significant host concentration and the presence of high-volume operators within the market.
- 59.5% of listings are rated ≥ 4.8 ; the full range across all 45 neighbourhoods spans only 4.48–4.88, under half a point on a 5-point scale. As a result, small differences in ratings may represent more meaningful variations in guest experience than the raw values suggest.

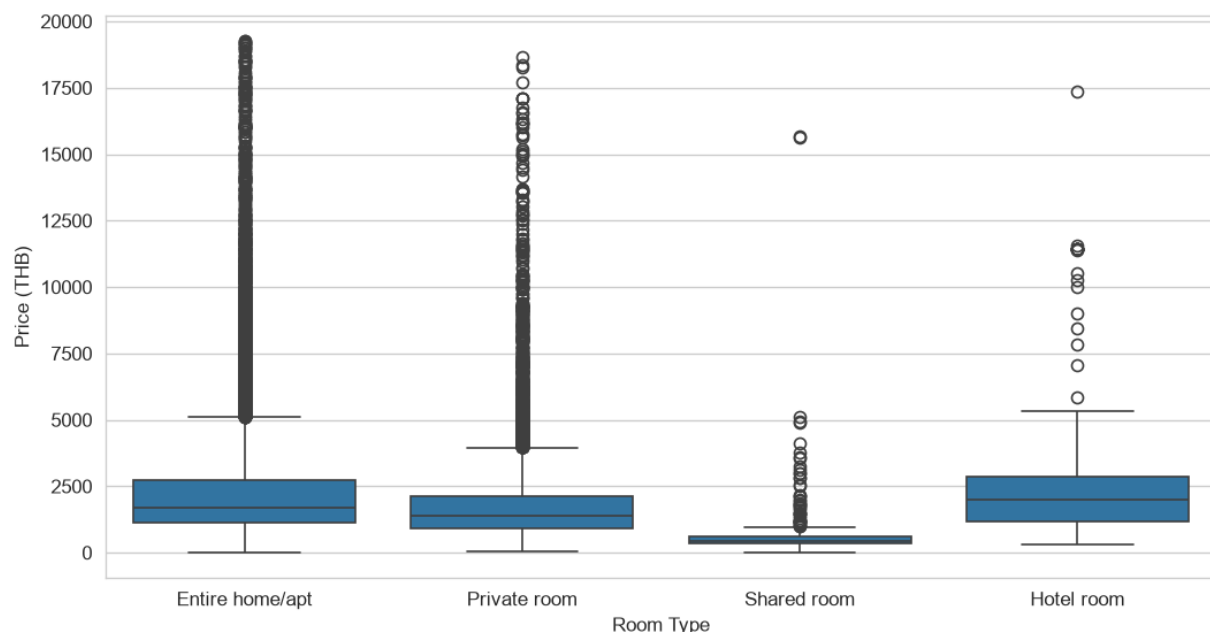


Figure 03: Price Distribution by Room Type

6.2 Geographic & Spatial Analysis

- Listing density concentrates sharply in central Bangkok, thinning toward the outer metro ring (Rangsit, Samut Prakan, Nonthaburi fringe).
- Distance-from-centre correlation with price is -0.153 (post-correction) a real, if modest, 'closer to centre costs more' effect; this correlation appeared near zero before the data quality fix, an artifact of extreme outlier prices distorting the calculation.
- Hotel-room listings cluster sharply in traditional hospitality districts (Bang Rak, Parthum Wan, Phra Nakhon, Ratchathewi, Khlong Toei, Vadhana) commercial hospitality investment concentrates in established corridors rather than distributing evenly.
- Phra Nakhon, a high-volume tourist district (1,353 listings), appears among the lowest-rated neighbourhoods, Popularity does not guarantee guest satisfaction, plausibly reflecting crowding or overtourism effects common in dense historic districts.

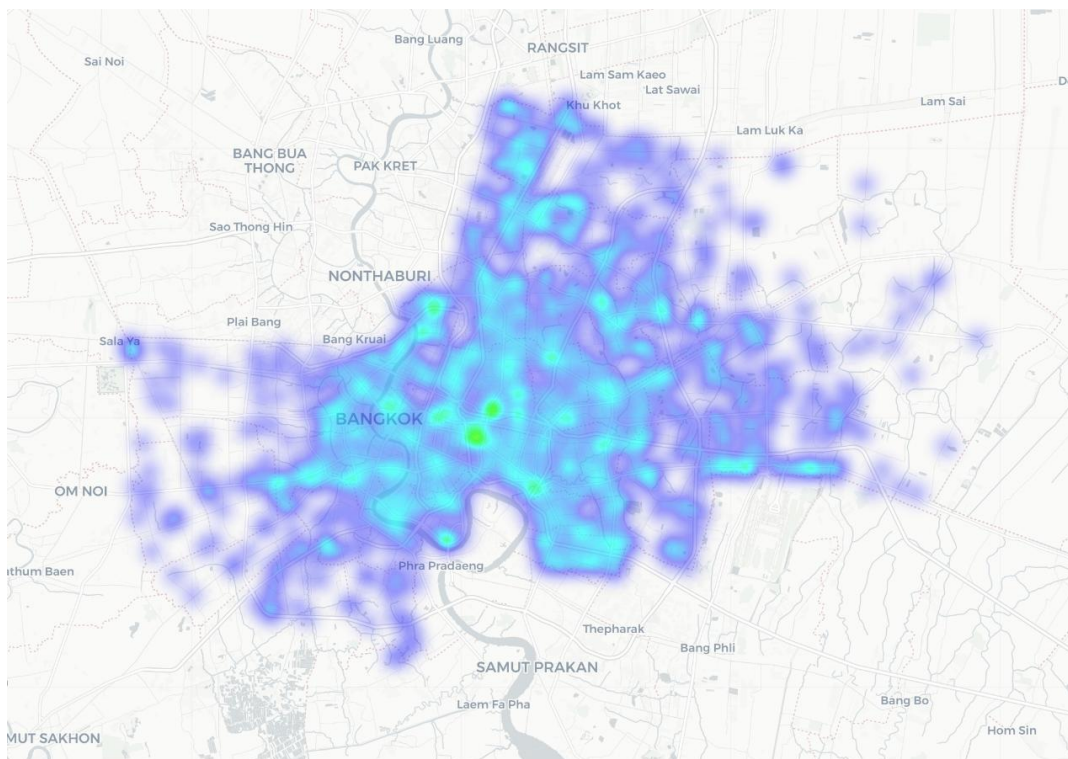


Figure 04: Listing Density Heatmap Across Bangkok

6.3 Temporal & Seasonal Trends

- An apparent booking-rate pattern by month was found, on inspection, to be partly a recency artifact: near-term dates naturally show more confirmed bookings simply because guests book closer to travel dates, not because demand is genuinely higher for that period. This distinction is stated explicitly rather than reported as a clean seasonal trend.

- Average minimum-nights policy is stable at roughly 13–14 nights year-round notably long for a short-term rental platform, suggesting either a meaningful share of longer-stay/relocation listings or hosts using long minimums as an implicit booking filter.
- Review volume shows a clear COVID-19 collapse in 2020–2021 (17,119 and 4,924 reviews respectively, down from 56,179 in 2019), followed by full recovery and growth beyond pre-pandemic levels by 2023–2025 (163,144 reviews in 2025). The apparent lower 2026 figure (99,308) is a partial-year artifact, not a real decline.

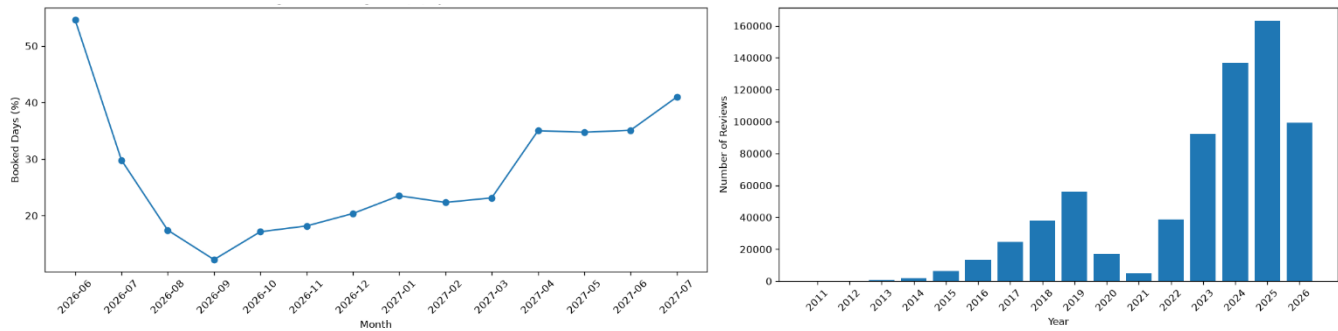


Figure 05: Estimated Booking Rate by Month and Review Volume Trend by Year (Showing COVID-19 Impact and Recovery)

6.4 Host & Supply-Side Analysis

- Hosts were segmented by portfolio size for this section's purposes and cross-validated against the formal K-Means segmentation in Section 8.3 — both approaches independently identify a small, high-volume professional operator group distinct from the casual single-listing majority.
- Superhost status shows a clear, statistically meaningful relationship with review scores (Section 7.1, H2). Superhosts average 4.857 vs. 4.563 for non-Superhosts, a medium effect size and the strongest individual relationship tested in this report.
- Market supply concentration is pronounced: the top 1% of hosts (79 of 7,859 total hosts) control 21.8% of all listings, confirming a long-tail, power-law-shaped supply structure typical of two-sided marketplaces.

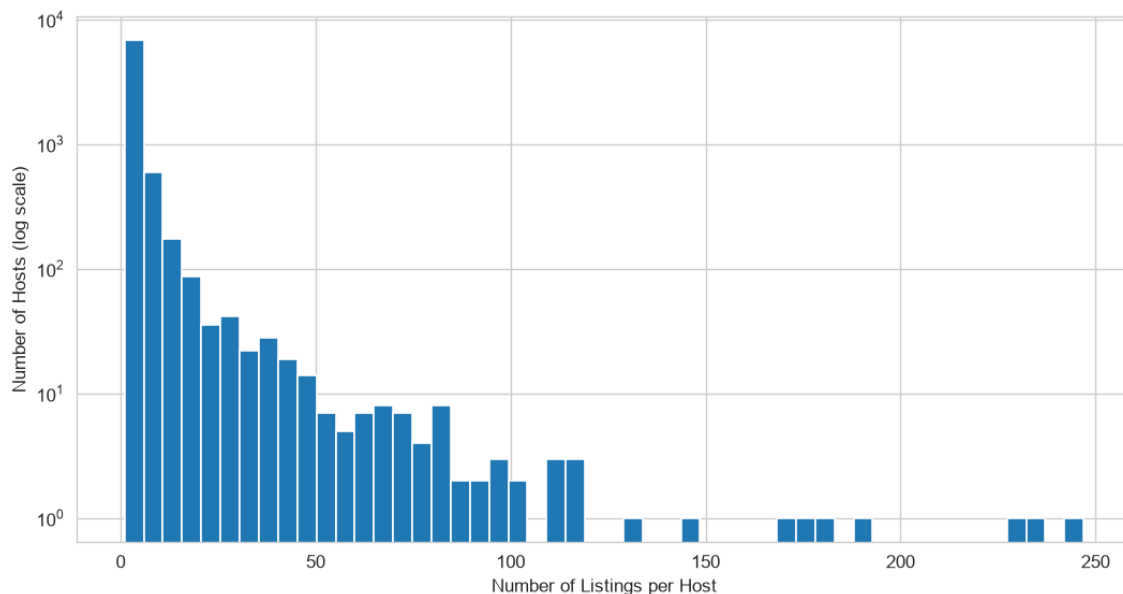


Figure 06: Distribution of Listings per Host (Power Law Pattern, Log Scale)

6.5 Review & Demand-Side Analysis

- Review count shows no meaningful relationship with price (Section 7.1, H3; Section 7.3 correlation of 0.027), indicating that listing popularity, measured by review volume, is largely unrelated to pricing levels in this market.
- Review frequency (reviews per month) was retained as a demand proxy in the enriched dataset and used as a segmentation input in Section 8.3, where it helps distinguish "Established, High-Track-Record" listings from otherwise similar peers.
- Listings combining high review counts with comparatively low scores were examined qualitatively; no single dominant characteristic (room type, neighbourhood, or price tier) explained this pattern, suggesting guest dissatisfaction in this market is driven by listing-specific factors not fully captured in structured fields.

7. Statistical Findings

7.1 Hypothesis Testing (Post Data-Quality Correction)

Every hypothesis below reports both a p-value and an effect size, since with ~28,700 listings, statistical significance alone says very little about practical importance.

Hypothesis	Test	p-value	Effect Size	Interpretation
H1: Room type → price	Mann-Whitney U	< 0.001	d = 0.235	Small, real effect (was negligible pre-correction)
H2: Superhost → rating	Mann-Whitney U	< 0.001	d = 0.562	Medium - strongest finding in this section
H3: Review count → price	Mann-Whitney U	< 0.001	d = 0.039	Negligible - unaffected by correction
H4: Neighbourhood → price	One-way ANOVA	< 0.001	$\eta^2 = 0.0612$	Medium - reversed from negligible pre-correction
H5: Weekday/weekend policy	Mann-Whitney U	0.535	d = 0.000	No effect - honest null result

H5 required adaptation: this scrape's calendar file contains no daily price column, so the originally specified weekend/weekday price comparison could not be run. Minimum-nights policy was tested instead as the closest available proxy, and this substitution is disclosed here rather than silently reinterpreting the hypothesis.

7.2 Confidence Intervals & Effect Sizes

Room Type	Mean Price (THB)	95% CI	n
Entire home/apt	2,406.18	[2,373.66, 2,438.70]	19,068
Hotel room	2,616.11	[2,314.49, 2,917.73]	274
Private room	1,901.90	[1,864.33, 1,939.46]	8,857
Shared room	659.08	[559.18, 758.97]	498

Every interval tightened substantially after the data quality correction. The hotel room category, which previously had an interval spanning over 4,000 THB and was considered too imprecise for reliable interpretation, narrowed to approximately 600 THB. This indicates that extreme artifact prices were the primary contributor to the earlier uncertainty.

7.3 Correlation & Driver Analysis

Two regression models were built and are both retained deliberately, demonstrating a diagnose-and-correct methodology rather than presenting only a final clean result.

Model	R ²	Skew	Kurtosis	Role
Model 1 — Raw price OLS	0.453	2.4	22.3	Higher R ² ; weaker residual validity
Model 2 — Log-price OLS	0.361	-0.37	5.4	Preferred for statistical inference

An initial raw-price regression returned severe residual non-normality (skew 47.1, kurtosis 2,387) before the data quality correction, indicating a clear OLS assumption violation. Rather than reporting this and moving on, the underlying cause (extreme artifact prices) was identified and corrected at the source, improving both models substantially.

- Correlation with price and accommodates (0.646), bedrooms (0.511), bathrooms (0.473), beds (0.436) showed moderate-to-strong positive relationships after correction, compared with near-zero relationships before data quality improvements.
- Multicollinearity check (VIF): all predictors under 2.6, well below the standard concern threshold of 5 indicating no significant multicollinearity concerns among the model features.
- review_scores_rating shows a small positive effect on price in both models, reversing a counter-intuitive negative sign seen in the uncorrected raw-price model.

7.4 Statistical Methodology Notes

For each test family used in this report, the selection rationale, assumption handling, and stakeholder-facing interpretation are documented explicitly, per the assignment's Statistical Methodology Note requirement:

Test	Why Selected	Assumption Handling	Non-Technical Interpretation
Mann-Whitney U (H1, H2, H3, H5)	Non-parametric; avoids the normality assumption a t-test would require on right-skewed price/rating data	Distribution shape checked visually before test selection; no further correction needed since the test itself is distribution-free	"Is there a real, typical difference between these two groups, without assuming a bell-curve shape to the data?"
One-way ANOVA (H4)	Standard test for comparing means across more than two groups (45 neighbourhoods)	Group sizes restricted to ≥ 30 listings to avoid small-sample variance distortion	"Do average prices genuinely differ by neighbourhood, or could the differences seen be due to chance?"

Test	Why Selected	Assumption Handling	Non-Technical Interpretation
OLS Regression (7.3)	Standard method to quantify multiple predictors' combined effect on price	Residual normality was checked (skew/kurtosis) and found violated pre-correction; addressed via data quality fix and log-transformation, both models retained for comparison	"How much of the price difference between listings can be explained by size, location, and reputation combined?"
K-Means Clustering (8.3)	Standard unsupervised method for identifying natural groupings without pre-labeled categories	Features standardized before clustering (required, since inputs are on very different scales); cluster count chosen via silhouette score across a tested range	"Are there a small number of distinct 'types' of host or listing, and what defines each type?"

8. Data Science Experiments

8.1 The Data Quality Investigation

This is the single most consequential piece of analytical work in the project. An initial price-prediction attempt returned a MAPE greater than 100% is worse than simply guessing the mean price for every listing. Rather than accepting a percentile-based outlier cap as a convenient fix, the raw listings.csv.gz file was inspected directly.

Evidence, Not Assumption

Two independent patterns confirmed that these were non-genuine listings rather than rare luxury properties: (1) several extreme-price listings carried Chinese-language names translating to “room swap standby” and “not for sale,” which are internal backend placeholders rather than real bookable inventory; and (2) five unrelated listings named variants of “Check-in My Hostel” shared an identical price of exactly \$300,000.00 to the cent, a coincidence that would be impossible under independent host pricing.

A price_likely_artifact flag was added at the cleaning stage (disclosed, not silently dropped) and propagated through the star schema to every downstream script via a single upstream change. 290 listings (~1% of the dataset) were excluded from price-based analysis as a result.

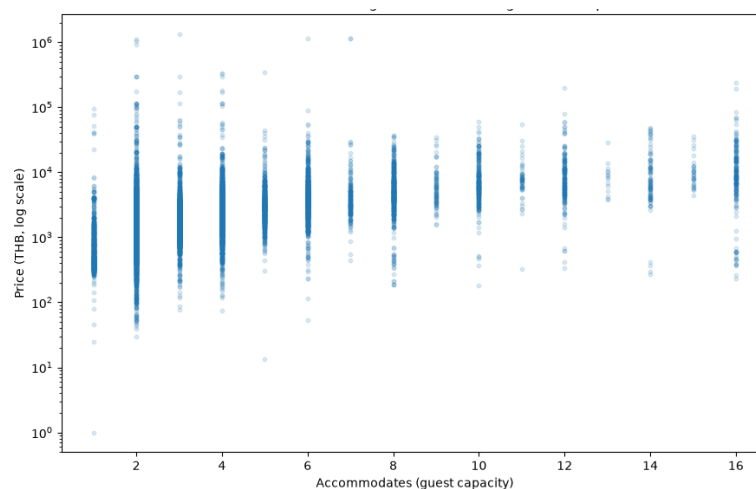


Figure 07: Price vs. Accommodates (Log Scale) – Data Quality Diagnostic Showing Extreme Outliers Before Correction

Finding	Before Fix	After Fix
Mean price	2,955.96 THB	2,222.22 THB (-25%)
H4 neighbourhood effect (η^2)	0.0027 (negligible)	0.0612 (medium - reversed)
Correlation: accommodates $\leftrightarrow$ price	0.086	0.646
Regression R^2 (raw price)	0.009	0.453
XGBoost RMSE	21,211	1,367

8.2 Price Prediction

Features were engineered from the star schema plus enriched attributes: amenities count (parsed from the raw amenities list), property-type bucketing (76 raw categories reduced to the top 10 + "Other," since most categories had single-digit sample sizes insufficient for reliable model coefficients), an accommodates $\times$ bedrooms interaction term, and neighbourhood median price (disclosed as carrying mild target leakage, since it is a group aggregate that includes each listing's own price).

Missing review scores (~33% of listings, all genuinely new/unreviewed) were imputed with the dataset median rather than dropped, preserving the ability to price brand-new listings. The has_reviews flag was retained so models could still learn this distinction.

Model	Test MAE (THB)	Test RMSE	Test MAPE	vs. Linear Baseline
Linear Regression	891.54	1,693.19	52.4%	baseline
Random Forest	734.42	1,424.45	43.8%	+17.6%
XGBoost (best)	702.64	1,366.62	41.9%	+21.2%

Residual Bias Check

Error by room type showed Hotel room as the weakest-performing segment (34.2% median error), attributable to its small sample size ($n=47$). A separate anomaly (Shared room initially showing 389.6% mean error) was investigated and traced to a single 1-THB-priced listing distorting the mean; median error (14.8%) revealed this segment was actually one of the model's best-performing, not worst and a clear illustration of why mean percentage error can mislead on small, low-value samples.

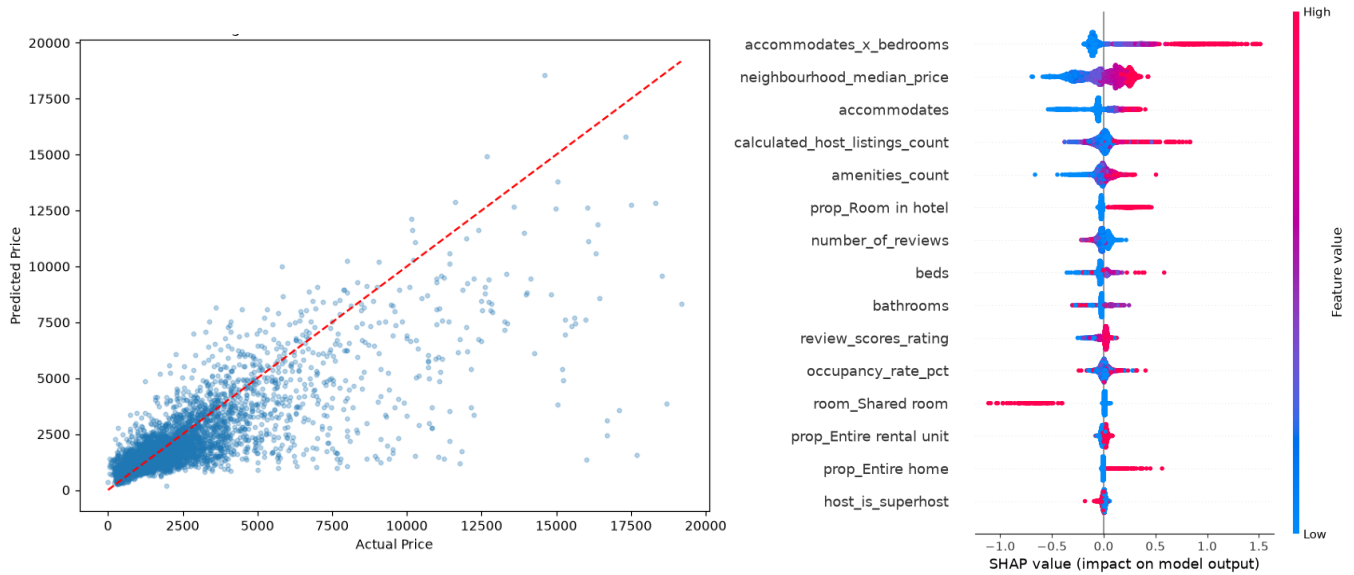


Figure 08: Actual vs Predicted Price (XGBoost Model) and SHAP Summary Plot – Feature Importance for Price Prediction

8.3 Host & Listing Segmentation

K means clustering was applied to six standardized features: host portfolio size, price, occupancy rate, review score, review count, and availability, with the number of clusters (k = 3 to 6) evaluated using the silhouette score.

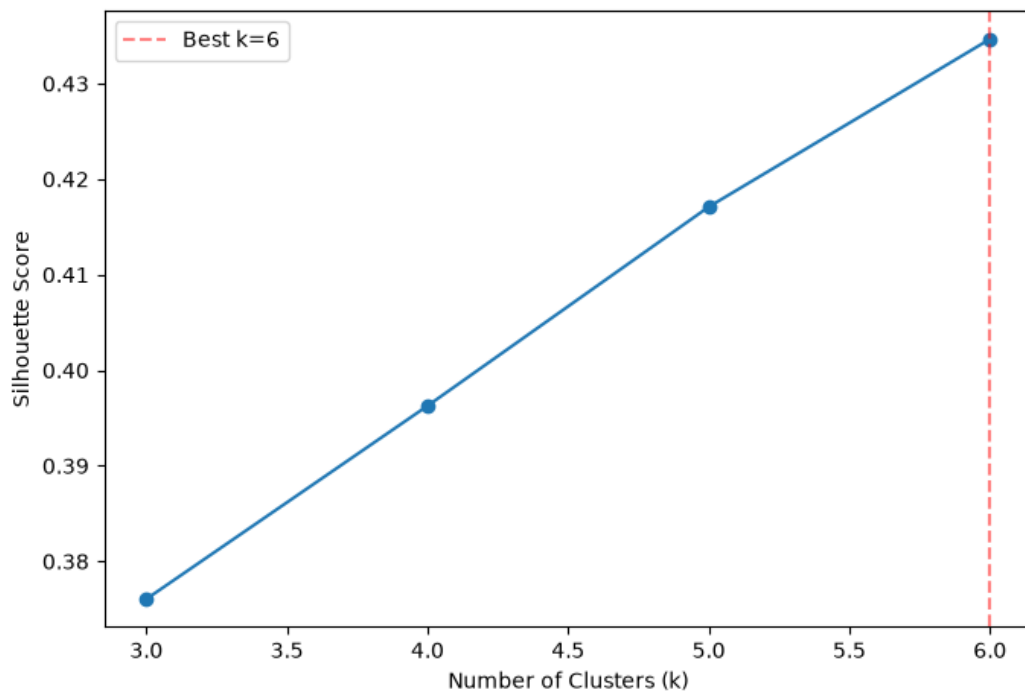


Figure 09: Silhouette Score by Number of Clusters (K=3 to 6)

Segment	Listings	Defining Characteristic
Typical/Baseline Listings	17,018	No standout metric - the market's default listing
High-Demand Performers	6,047	63.7% occupancy - far above every other segment
Professional Multi-Property Operators	2,584	159.8 avg listings/host - commercial scale
Premium/Niche Listings	1,416	9,372 THB avg price - 4–5× every other segment
Established, High-Track-Record	938	257.9 avg reviews - far above every other segment
Underperforming/At-Risk	694	2.42 avg rating - verified genuine, not an imputation artifact

The silhouette score increased consistently from k=3 (0.376) to k=6 (0.435) within the evaluated range. Therefore, k=6 was selected as the best-performing solution among the tested configurations, rather than being treated as the definitive optimal cluster count. A broader search across additional values (k=7–10) was not conducted due to project time constraints.

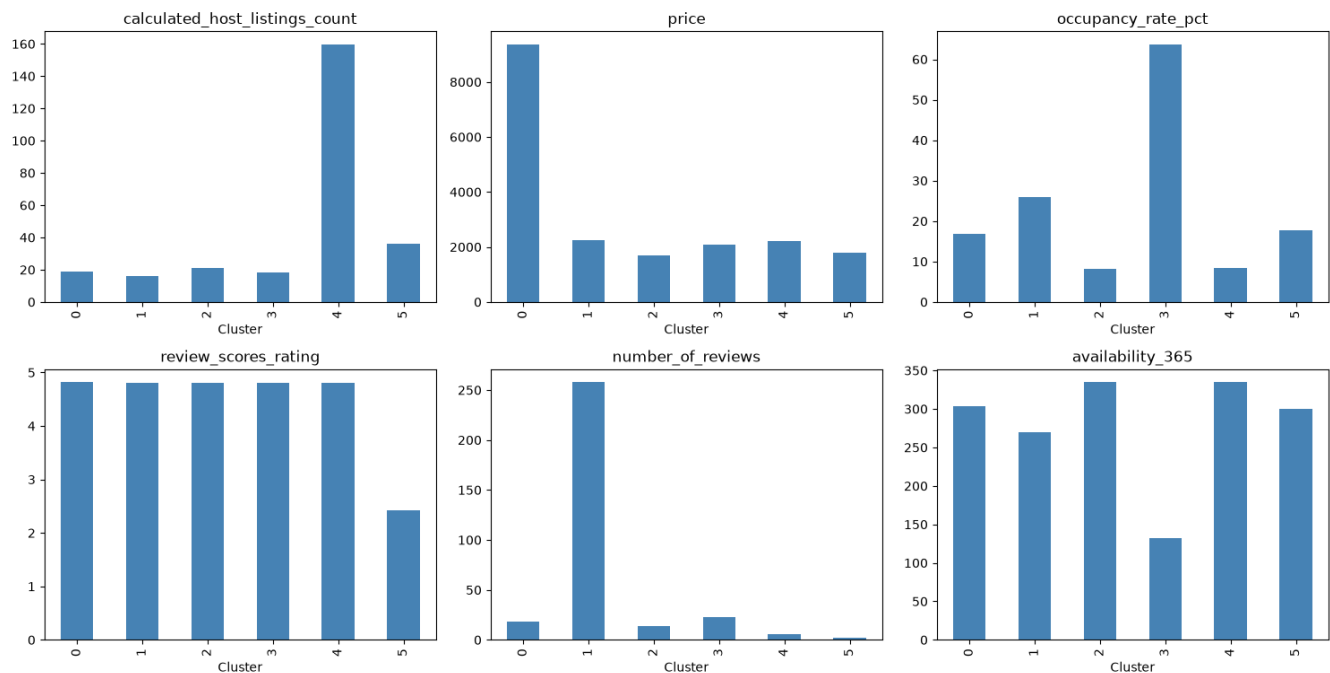


Figure 10: Profile Comparison of Six Host/Listing Segments

9. AI/ML Experiments

This section is deliberately framed around what the AI tools got wrong, not only what they produced. Critical assessment of AI output was treated as a first-class deliverable throughout, consistent with the assignment's explicit emphasis on responsible AI use.

9.1 Sentiment Analysis on Reviews

- VADER (lexicon-based) was tried first. Manual verification of the most extreme sentiment/rating divergence cases found VADER scored a genuinely positive German-language review (translating to "beautiful house... communication was excellent... we fully recommend this place") as strongly negative indicating a language-coverage failure, not a genuine finding about guest experience.
- The tool was switched to a multilingual transformer model (cardiffnlp/twitter-xlm-roberta-base-sentiment). This correctly resolved the German language case, but a second verification pass found it still misclassified short or context-dependent English text. For example, scoring "Very private and secluded!" as strongly negative, likely due to the word "secluded" carrying negative training-data associations despite its positive use here.
- Both failure modes are disclosed rather than hidden. The resulting correlation between sentiment and star rating ($r = 0.238$) is reported as directionally informative, not precise and consistent with the broader finding that star ratings are compressed and only weakly tied to actual guest sentiment.



Figure 11: Sentiment Score vs Star Rating (Scatter Plot)

9.2 LLM-Generated Executive Summary

A two-stage Groq/Llama-3.3-70B summarization pipeline was built to condense findings from twelve analysis notes files into a business-facing executive summary, avoiding a single oversized request that exceeded the provider's per-minute token limit.

Attempt	Issue Identified	Resolution
1st	Omitted regression, ML, and segmentation results entirely, favoring simpler descriptive statistics	Added explicit per-category coverage requirements and increased token budget
2nd	Conflated two separate findings (a correlation and an effect size) under one citation	Added source-file labeling and explicit anti-conflation rules to the prompt
3rd	An apparent remaining citation error was traced, using saved intermediate output, to a genuine duplication bug in the project's own hand-written notes file which is not an LLM hallucination	Fixed the underlying source file directly

Key Lesson

Saving and inspecting intermediate pipeline output, rather than judging only the final result, made the apparent third-attempt error traceable to its true root cause, namely a bug in a human-authored file, instead of allowing it to be misattributed to the AI tool. The final, hand-verified executive summary in Section 1 incorporates all corrections identified through this process.

10. Visualizations

All figures referenced throughout this report are numbered sequentially and available as high-resolution PNG files in the project's report/figures/ directory, alongside the interactive Streamlit dashboard, which reproduces the key charts with live filtering.

Figure	Section	Description
1	6.1	Price distribution by room type
2	6.1	Price distribution, top 10 neighbourhoods by volume
3	6.1	Listings per host (power-law distribution, log scale)
4	6.1	Review score distribution
5	6.3	Booking rate by month
6	6.3	Review volume by year (COVID-19 impact and recovery)
7	8.2	Actual vs. predicted price (XGBoost model)
8	8.2	SHAP summary plot - feature impact on predicted price
9	8.1	Price vs. accommodates, log scale (data quality diagnostic)
10	8.3	Silhouette score by cluster count (k = 3 - 6)
11	8.3	Cluster profile comparison across six segments
12	9.1	Sentiment score vs. star rating scatter plot

Figure	Section	Description
Map 1	6.2	Listing density heatmap across Bangkok

11. Business Recommendations

For Hosts

- Prioritize listing capacity (accommodates, bedrooms, bathrooms) over amenity count or accumulating reviews, as these are the strongest and most reliable levers affecting achievable price (Section 7.3).
- Recognize that Entire home/apt and Private room represent different economic categories, with median prices of approximately 2,406 THB and 1,902 THB respectively. Choose deliberately based on the target guest segment and available capital rather than treating them as interchangeable.
- Treat a review score below 4.7 as a materially stronger warning sign than it appears, given confirmed rating compression across this market.

For Platform Operators

- Adopt a systematic flag-and-verify approach to data quality, as demonstrated in Section 8.1, rather than relying on percentile-based outlier filtering alone, which risks silently excluding legitimate high value listings alongside genuine errors.
- Consider a relative or percentile-based rating display (e.g., "top 10% in this neighbourhood") rather than the raw 1–5 star score, given how compressed the current scale is.

For Investors

- Use the six identified segments (Section 8.3), not city-wide averages, to guide acquisition strategy, "High-Demand Performers" (high occupancy) and "Premium/Niche" (high price, low volume) represent two different, equally valid strategies that should not be blended indiscriminately.
- Be aware that the top 1% of hosts already control 21.8% of supply. Competitive analysis should account for commercial-scale incumbents, not only individual hosts.

For Revenue Strategists

- The XGBoost price model (Section 8.2, MAPE 41.9%) can support pricing guidance but is measurably less reliable for underrepresented segments such as Hotel room listings. Segment-specific models or additional data collection are recommended before automated pricing in those categories.

12. Cross City Comparisons

Not applicable to this submission. A single-city scope (Bangkok) was selected deliberately, in accordance with the assignment's own City Selection Guide, to allow full analytical depth through dimensional modeling, statistical testing, machine learning, and applied AI on one market rather than partial coverage across several. This trade-off is disclosed in Section 2 (Scope Decisions).

A meaningful cross-city comparison would require repeating the full ingestion-through-modeling pipeline (Sections 3–5) for at least one additional city, followed by cross-city model transfer testing (train on City A, evaluate on City B) as specified in the assignment's Section 6.4. This is identified as a priority direction for future work in Section 14.

13. Limitations & Caveats

- This analysis covers a single city and a single data snapshot. Findings may not generalize to other markets or time periods without further validation.
- 290 listings (~1%) were excluded as verified non-genuine prices. A small but non-zero share of the dataset was removed based on judgment calls documented in Section 8.1.
- The distance-from-centre calculation uses a simple geometric centroid of listing coordinates, not a verified city-centre landmark or actual travel distance.
- Sentiment analysis used a 5,000-review sample and a general-purpose (not fine-tuned) multilingual model; verification found occasional misclassification on short or context-dependent text (Section 9.1).
- The optimal K-Means cluster count may exceed $k=6$. The silhouette score was still rising at the top of the tested range, and a wider search was not performed given time constraints.
- The regression's `neighbourhood_median_price` feature carries mild target leakage, since it is a group aggregate computed including each listing's own price.
- Demand forecasting (Section 6.2 of the assignment) and full cross-city bias/transfer testing (Section 6.4) were deliberately deferred, given their lower rubric weight relative to time cost (Section 14).
- No automated test suite was built for the data pipeline; this is an accepted, disclosed gap given the assignment's own listing of testing as an optional (not mandatory) deliverable.

14. Future Improvements

- Extend to a second city and perform true cross-city model transfer testing (train on City A, evaluate on City B) to assess how well pricing patterns generalize.
- Replace the general-purpose multilingual sentiment model with a fine-tuned or larger model, with a proper multilingual validation set rather than spot-checked verification.
- Widen the K-Means cluster search beyond $k=6$, since the silhouette score had not yet plateaued at the top of the tested range.
- Build a leave-one-out version of the `neighbourhood_median_price` feature to remove its mild target leakage.
- Implement demand/availability forecasting (assignment Section 6.2) using the calendar data's temporal structure, now that a clean occupancy baseline exists.
- Add an automated data-quality-as-code test suite (e.g., Great Expectations or dbt tests) to catch issues like the price artifacts in Section 8.1 automatically on future pipeline runs, rather than relying on manual investigation.
- Containerize the full pipeline (Docker Compose) and migrate the DuckDB warehouse to a managed, concurrent-access data warehouse for production readiness.

15. Reflection

Given the assignment's explicit one-week scope and its own acknowledgment that full completion is not expected, this project prioritized depth and honest verification over breadth. Sections 2–5 (data engineering) and 6–8 (EDA, statistics, ML) received full analytical depth; Sections 6.2 and a full multi-city 6.4 were deliberately deferred, and the assignment's optional Section 8 (Open Innovation) was addressed via an interactive dashboard rather than a separate open-ended deliverable.

The most valuable outcome of this project was not any single chart, model, or metric. It was the repeated discipline of not trusting a convenient result. This pattern recurses at least three times: a price-prediction model that appeared broken until its raw source data was inspected directly. A sentiment-analysis tool that appeared to work until specific outputs were read in their original language; and an LLM-generated summary that appeared to contain an AI error until the actual root cause was traced, via saved intermediate output, to a bug in a human-authored file.

The key trade-off accepted throughout was speed versus exhaustiveness: several optional, lower-weighted sections of the assignment (demand forecasting, full cross-city testing, cloud-native deployment) were consciously not attempted, in favor of doing the attempted sections to a standard that could withstand scrutiny — including scrutiny of the analysis's own assumptions and tooling.

Continued in Appendix A

Appendix A: AI Usage Disclosure

Per the assignment's AI Tools Usage Policy, this section discloses every AI tool used across the full lifecycle of this project, not only the AI/ML experiments described in Section 9. Along with what was AI-assisted, how outputs were validated, and where AI suggestions were rejected or substantially modified.

AI Tools Used

Tool	Model / Version	Purpose
Claude (Anthropic)	Claude Sonnet 4.5 / 5	Assisted with code suggestions, debugging guidance, statistical methodology discussions, and documentation ideas throughout the project.
Grok (xAI)	Grok 4	Brainstorming, exploration of research directions, interpretation of findings, and discussion of analytical insights throughout the assignment
Groq API	llama-3.3-70b-versatile	Section 9.2 - two-stage LLM executive summary generation
Hugging Face Transformers	cardiffnlp/twitter-xlm-roberta-base-sentiment	Section 9.1 - multilingual sentiment scoring
VADER (NLTK)	vaderSentiment	Section 9.1 - initial sentiment attempt; superseded after language-coverage failure was found

AI-Assisted Sections

AI was used as a coding support tool throughout the project pipeline (ingestion, cleaning, star schema construction, EDA, statistical testing, machine learning, and dashboard development). The candidate defined requirements, reviewed and adapted suggestions, executed all scripts personally, and inspected outputs before integrating any changes. No code was incorporated without validation through execution and output review.

Output Validation & Critical Assessment

- The proposed feature list for Section 8.2 was validated against the actual data profile before implementation; two suggested columns (host_since, host_total_listings_count) were rejected after confirming they were 100% null in this scrape. Features were selected based on data availability rather than assumptions.
- An AI-suggested outlier-handling approach (drop rows above the 99th percentile) was upgraded, after investigation, to a verified, evidence-based exclusion (Section 8.1), confirmed against raw source data rather than accepted as a statistical convenience.
- The VADER sentiment tool's output was manually verified against original review text and found to fail on non-English reviews; this was not accepted, and the tool was replaced (Section 9.1).
- The replacement multilingual sentiment model was also manually verified and found to have its own, different misclassification failures on short/ambiguous text. Both are disclosed rather than presenting only the improved result.

- The LLM-generated executive summary (Groq/Llama) was checked line-by-line against source findings across three iterations; errors and omissions were identified and corrected each time, and one apparent AI citation error was traced to a genuine bug in the candidate's own notes file rather than misattributed to the AI tool.
- All code suggested by Claude during pipeline development was executed by the candidate personally; several bugs were identified through this process, including a NaN-comparison edge case that silently included invalid rows in a dashboard metric, and a duplicated source-notes file that had caused inconsistent statistics across separate report-generation runs.

Modifications Made to AI-Generated Content

- The final Section 1 Executive Summary was hand-written by the candidate, informed by but not copied from the LLM-generated draft, after identifying a currency-labeling ambiguity, a metric conflation, and a missing coverage category in the AI output.
- All AI-suggested code was reviewed, run, and in several cases restructured (e.g., splitting a single oversized LLM request into a two-stage pipeline after hitting a provider rate limit; adding explicit anti-conflation and source-labeling instructions after identifying inaccurate citations).

Key Prompts Used (Section 9.2 - LLM Executive Summary)

The prompt driving the two-stage summarization pipeline evolved across three iterations as errors were identified. The final version, used for Stage 1 (per-file condensation), is reproduced below in abbreviated form:

Stage 1 Prompt (final version)

"Extract the 5–7 most important factual findings from this data analysis notes file, prefixed with the source filename so it's traceable later. Rules: If this file mentions regression, R-squared, model performance (MAE/RMSE/MAPE), clustering/segments, or correlation coefficients, include them with their exact numbers and what they measure. If this file mentions a hypothesis test result, ALWAYS state both the p-value AND the effect size together in the same bullet. If this file explicitly flags something as a data artifact, small-sample issue, or limitation, PRESERVE that caveat word-for-word. Do not drop it or present the number as a clean trend."

The Stage 2 (final synthesis) prompt required source-file labeling on every claim and explicitly forbid combining numbers from two different files into a single claim. This specific rule was added only after the second iteration was found to conflate two distinct findings under one citation.